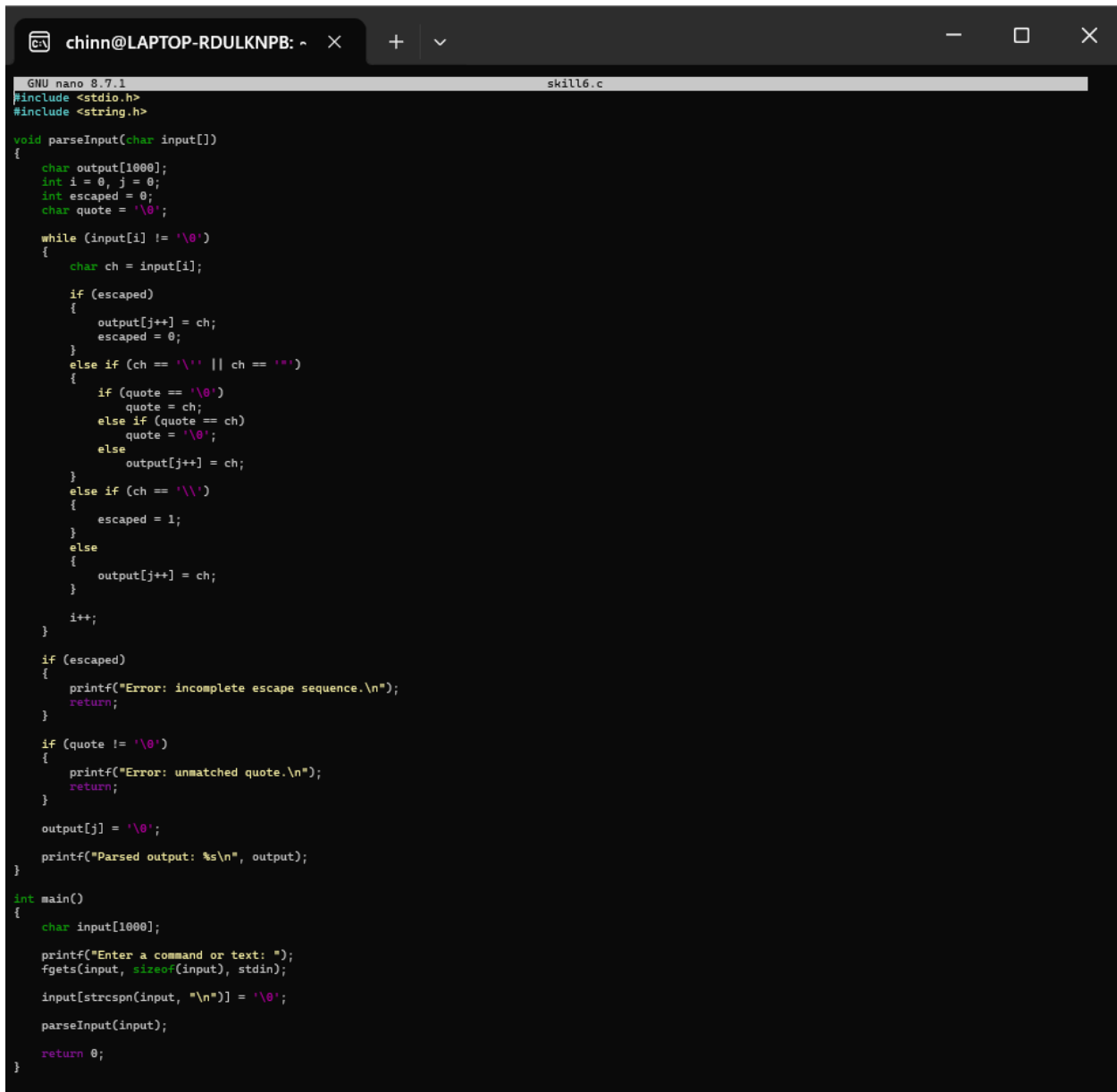


Name: V.Siri Chandana
Roll No:2520030609
Section:2
Subject:OS

Skill 6 Programs

1.

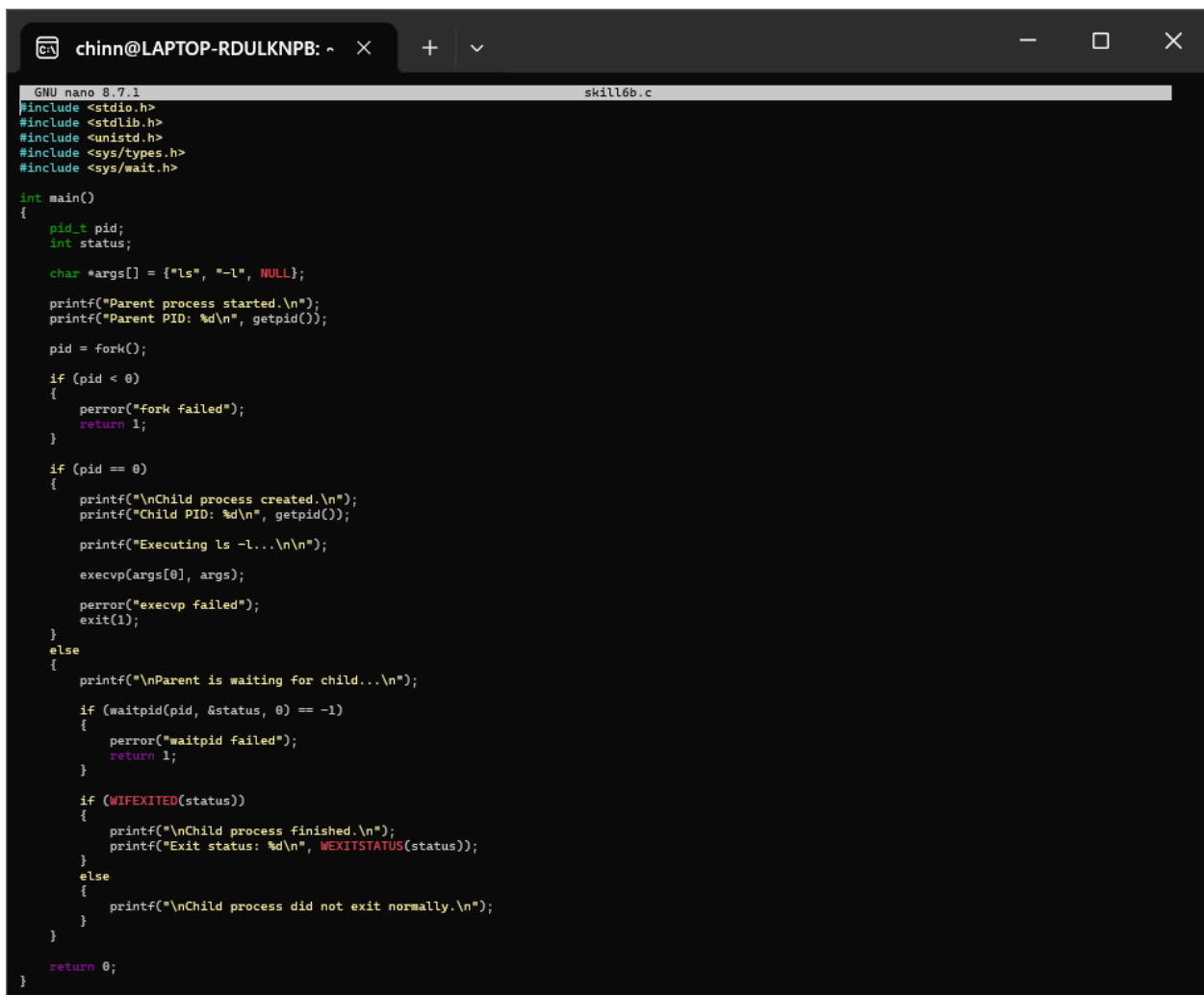


```
chinn@LAPTOP-RDULKNPB: ~  
GNU nano 8.7.1 skill6.c  
#include <stdio.h>  
#include <string.h>  
  
void parseInput(char input[])  
{  
    char output[1000];  
    int i = 0, j = 0;  
    int escaped = 0;  
    char quote = '\\0';  
  
    while (input[i] != '\\0')  
    {  
        char ch = input[i];  
  
        if (escaped)  
        {  
            output[j++] = ch;  
            escaped = 0;  
        }  
        else if (ch == '\\' || ch == '\"')  
        {  
            if (quote == '\\0')  
                quote = ch;  
            else if (quote == ch)  
                quote = '\\0';  
            else  
                output[j++] = ch;  
        }  
        else if (ch == '\\\\')  
        {  
            escaped = 1;  
        }  
        else  
        {  
            output[j++] = ch;  
        }  
  
        i++;  
    }  
  
    if (escaped)  
    {  
        printf("Error: incomplete escape sequence.\\n");  
        return;  
    }  
  
    if (quote != '\\0')  
    {  
        printf("Error: unmatched quote.\\n");  
        return;  
    }  
  
    output[j] = '\\0';  
    printf("Parsed output: %s\\n", output);  
}  
  
int main()  
{  
    char input[1000];  
  
    printf("Enter a command or text: ");  
    fgets(input, sizeof(input), stdin);  
  
    input[strcspn(input, "\\n")] = '\\0';  
    parseInput(input);  
  
    return 0;  
}
```

Output:

```
chinn@LAPTOP-RDULKNPB:~/OS_SKILL$ nano skill6.c
chinn@LAPTOP-RDULKNPB:~/OS_SKILL$ gcc skill6.c -o skill6
chinn@LAPTOP-RDULKNPB:~/OS_SKILL$ ./skill6
Enter a command or text: can you tell me correct
Parsed output: can you tell me correct
```

2.



```
chinn@LAPTOP-RDULKNPB: ~ × + ▾ - □ ×
GNU nano 8.7.1 skill6b.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;
    int status;

    char *args[] = {"ls", "-l", NULL};

    printf("Parent process started.\n");
    printf("Parent PID: %d\n", getpid());

    pid = fork();

    if (pid < 0)
    {
        perror("fork failed");
        return 1;
    }

    if (pid == 0)
    {
        printf("\nChild process created.\n");
        printf("Child PID: %d\n", getpid());

        printf("Executing ls -l...\n");

        execvp(args[0], args);

        perror("execvp failed");
        exit(1);
    }
    else
    {
        printf("\nParent is waiting for child...\n");

        if (waitpid(pid, &status, 0) == -1)
        {
            perror("waitpid failed");
            return 1;
        }

        if (WIFEXITED(status))
        {
            printf("\nChild process finished.\n");
            printf("Exit status: %d\n", WEXITSTATUS(status));
        }
        else
        {
            printf("\nChild process did not exit normally.\n");
        }
    }

    return 0;
}
```

Output:

```
chinn@LAPTOP-RDULKNPB:~/OS_SKILL$ nano skill6b.c
chinn@LAPTOP-RDULKNPB:~/OS_SKILL$ gcc skill6b.c -o skill6b
chinn@LAPTOP-RDULKNPB:~/OS_SKILL$ ./skill6b
Parent process started.
Parent PID: 691

Parent is waiting for child...

Child process created.
Child PID: 692
Executing ls -l...

total 204
-rwxr-xr-x 1 chinn chinn 16072 Sep 30 17:17 practical6
-rw-r--r-- 1 chinn chinn 391 Oct 1 12:57 practical6.c
-rwxr-xr-x 1 chinn chinn 16160 Sep 30 16:40 practical7
-rw-r--r-- 1 chinn chinn 443 Sep 30 16:39 practical7.c
-rwxr-xr-x 1 chinn chinn 16088 Sep 30 16:43 practical8
-rw-r--r-- 1 chinn chinn 240 Sep 30 16:43 practical8.c
-rwxr-xr-x 1 chinn chinn 16040 Sep 30 16:49 practical8b
-rw-r--r-- 1 chinn chinn 299 Sep 30 16:48 practical8b.c
-rwxr-xr-x 1 chinn chinn 16208 Aug 13 05:02 program1
-rw-r--r-- 1 chinn chinn 532 Aug 13 05:02 program1.c
-rwxr-xr-x 1 chinn chinn 16456 Aug 13 05:36 program2
-rw-r--r-- 1 chinn chinn 976 Aug 13 05:36 program2.c
-rwxr-xr-x 1 chinn chinn 16944 Sep 29 23:09 shellforge
-rw-r--r-- 1 chinn chinn 6650 Sep 29 23:09 shellforge.c
-rw-r--r-- 1 chinn chinn 6649 Sep 29 23:08 shellforge.c.save
-rwxr-xr-x 1 chinn chinn 16208 Oct 1 12:58 skill6
-rw-r--r-- 1 chinn chinn 1183 Oct 1 12:57 skill6.c
-rwxr-xr-x 1 chinn chinn 16304 Oct 1 13:01 skill6b
-rw-r--r-- 1 chinn chinn 1098 Oct 1 13:01 skill6b.c
drwxr-xr-x 2 chinn chinn 4096 Sep 29 23:01 temp

Child process finished.
Exit status: 0
chinn@LAPTOP-RDULKNPB:~/OS_SKILL$ |
```